Parameters in a CNN (Example 1)

```python
my_input = Input(shape = (10, 10, 1))
my_output = Convolution2D(1, 3)(my_input)
my_output = Activation('sigmoid')(my_output)
model = Model(my_input, my_output)
```

```python
model.summary()
```

Layer (type)	Output Shape	Param #
input_17 (InputLayer)	[(None, 10, 10, 1)]	0
conv2d_19 (Conv2D)	(None, 8, 8, 1)	10
activation_19 (Activation)	(None, 8, 8, 1)	0

Total params: 10
Trainable params: 10
Non-trainable params: 0

```python
my_input = Input(shape = (10, 10, 1))
my_output = Convolution2D(5, 3)(my_input)
my_output = Activation('sigmoid')(my_output)
my_output = Convolution2D(2, 3)(my_output)
my_output = Activation('sigmoid')(my_output)
model = Model(my_input, my_output)
```

```python
model.summary()
```

Layer (type)	Output Shape	Param #
input_19 (InputLayer)	[(None, 10, 10, 1)]	0
conv2d_22 (Conv2D)	(None, 8, 8, 5)	50
activation_22 (Activation)	(None, 8, 8, 5)	0
conv2d_23 (Conv2D)	(None, 6, 6, 2)	92
activation_23 (Activation)	(None, 6, 6, 2)	0

Total params: 142
Trainable params: 142
Non-trainable params: 0

Parameters in a CNN (Example 3)

```
my_input = Input(shape = (100, 100, 3))
my_output = Convolution2D(8, 3)(my_input)
my_output = Activation('sigmoid')(my_output)
my_output = Convolution2D(1, 3)(my_output)
my_output = Activation('sigmoid')(my_output)
model = Model(my_input, my_output)
```

```
model.summary()
```

Layer (type)	Output Shape	Param #
input_20 (InputLayer)	[(None, 100, 100, 3)]	0
conv2d_24 (Conv2D)	(None, 98, 98, 8)	224
activation_24 (Activation)	(None, 98, 98, 8)	0
conv2d_25 (Conv2D)	(None, 96, 96, 1)	73
activation_25 (Activation)	(None, 96, 96, 1)	0

Total params: 297
Trainable params: 297
Non-trainable params: 0

Parameters in a CNN (Example 4)

```
my_input = Input(shape = (100, 5))
my_output = Convolution1D(8, 3)(my_input)
my_output = Activation('sigmoid')(my_output)
my_output = Convolution1D(1, 3)(my_output)
my_output = Activation('sigmoid')(my_output)
model = Model(my_input, my_output)
```

```
model.summary()
```

Layer (type)	Output Shape	Param #
input_25 (InputLayer)	[(None, 100, 5)]	0
conv1d_4 (Conv1D)	(None, 98, 8)	128
activation_30 (Activation)	(None, 98, 8)	0
conv1d_5 (Conv1D)	(None, 96, 1)	25
activation_31 (Activation)	(None, 96, 1)	0

Total params: 153
Trainable params: 153
Non-trainable params: 0